

# Determining the Activity of a $^{90}\text{Sr}$ Source via Systematic Corrections to the Inverse-Square-Scaled Count Rate in the Near and Far Field

Sammy Chen

**Abstract**—We measured the count rate from a  $^{90}\text{Sr}$  beta source with a Si detector over separations 0-90 cm and analysed the inverse-square-scaled quantity  $u(d) = R(d)d^2$ , which should be constant for an ideal point source and point detector. Deviations from constancy were observed at small  $d$  (offset distance, detector solid angle and paralyzable dead-time) and at large  $d$  (air energy attenuation). We modelled the true rate as  $R(d) = \frac{A\eta}{4\pi} \Omega_{\text{rect}}(d + d_0) T_{\text{air}}(d + d_0)$ , and related the measured rate by  $R_{\text{meas}} = R e^{-R\tau}$ , with  $\tau = 3.12 \pm 0.06 \mu\text{s}$  estimated from oscilloscope pulse. Fitting yielded  $d_0 = 1.336 \pm 0.075 \text{ cm}$  and an effective low-energy spectral weight  $f_{\text{low}} = 0.169 \pm 0.016$ , with good agreement ( $p \simeq 0.87$ ); background count  $0.12 \pm 0.01 \text{ s}^{-1}$  was negligible. Adding an air-transmission term decreases  $\chi^2_{\text{red}}$  from  $\approx 35$  to  $\approx 2.5$ , with  $\Delta\text{AIC} \approx 178$  relative to the next-best model.  $u(d)$  is consistent with a constant  $K_{\text{est}} = (1.6128 \pm 0.0058) \times 10^5 \text{ cps} \cdot \text{cm}^2$ , implying  $A\eta = (2.027 \pm 0.007) \times 10^6 \text{ Bq}$  and  $\eta \approx 0.405$  relative to the 5 MBq source label.

## I. INTRODUCTION

For an isotropic point source and a point detector, the measured count rate is expected to scale as  $R(d) \propto d^{-2}$  with source-detector separation  $d$ . It is therefore convenient to analyse the inverse-square-scaled quantity  $u(d) = R(d)d^2$ , which should be constant if geometry is the only effect. In practice, our  $^{90}\text{Sr}$  data show systematic deviations from constancy at both small and large separations, indicating additional physical and instrumental effects beyond the ideal model. In this work, we account for these deviations by including a distance offset, the finite rectangular detector solid angle, paralyzable dead-time losses, background subtraction, and an effective air-transmission term. The corrected model is then tested for consistency and used to estimate the source strength and infer the  $^{90}\text{Sr}$  activity.

## II. THEORY

### A. Including rectangular solid angle

For an isotropic  $^{90}\text{Sr}$  source with activity  $A$  ( $\text{decays s}^{-1}$ ) and overall intrinsic detector efficiency  $\eta$  at separation  $d$ , the point-detector approximation gives

$$R(d) = \frac{n}{\Delta t} \approx \frac{A\eta A_{\text{det}}}{4\pi d^2}. \quad (1)$$

Defining the inverse-square-scaled quantity<sup>2</sup>

$$u(d) = R(d)d^2, \quad (2)$$

we obtain a constant in the far field,

$$u(d) \approx \frac{A\eta A_{\text{det}}}{4\pi} = K, \quad (3)$$

where  $K$  is the inverse-square-scaled count rate constant. At short distances, the approximation in Eq.1 breaks down, and the detector cannot be treated as a point. The correct geometric acceptance is the rectangular solid angle  $\Omega_{\text{rect}}(d)$ <sup>3</sup> subtended by the  $1\text{cm} \times 1\text{cm}$  detector. Including a distance offset  $d_0$ , we can define the actual separation distance as  $r = d + d_0$ , the near-field model becomes

$$u(d) \approx \frac{A\eta}{4\pi} \Omega_{\text{rect}}(d + d_0) d^2, \quad (4)$$

which reduces to  $u \rightarrow \text{constant } K$  in the far field ( $d \gg a, b$ ) since  $\Omega_{\text{rect}} \approx A_{\text{det}}/r^2$ .

### B. Air attenuation correction

Air acts like a very thin absorber whose areal density increases with distance  $\sigma_{\text{air}}(r) = \rho_{\text{air}} r$ . We can treat a  $\beta$  as detected only if its initial kinetic energy exceeds the minimum energy required to travel through  $\sigma_{\text{air}}$ . Using the lab script's range energy relation,<sup>2</sup> we can get

$$E_{\text{min}}(d) = \sqrt{\left( \frac{(1 + \sigma_{\text{air}}(d)/0.11)^2 - 1}{22.4} \right)}. \quad (5)$$

We can define the transmission rate  $T(r)$  as the surviving fraction of the  $\beta$  spectrum<sup>4</sup> above the  $E_{\text{min}}$ . Considering both  $\beta$  branches (endpoints:  $E_{\text{max}} = 0.54 \text{ MeV}$  and  $2.27 \text{ MeV}$ ) with an effective weight  $f_{\text{low}}$ , the air transmission rate is

$$T_{\text{air}}(d) = f_{\text{low}} T_{0.54}(d) + (1 - f_{\text{low}}) T_{2.27}(d). \quad (6)$$

### C. Dead time correction

The detector cannot register a second event until the signal recovers below the threshold. Therefore, the dead time  $\tau$  is the minimum resolvable separation between events. For a paralyzable detector,<sup>5</sup> the measured rate  $R_{\text{meas}}$  and the true rate  $R$  are related

$$R_{\text{meas}}(d) = R(d) e^{-R(d)\tau}. \quad (7)$$

### D. Complete model combining all systematic uncertainties

Combining all the above, we can define

$$R(d) = \frac{A\eta}{4\pi} \Omega_{\text{rect}}(d + d_0) T(d + d_0). \quad (8)$$

Furthermore, the full model for  $u_{\text{meas}}(d)$

$$u_{\text{meas}}(d) = d^2 R e^{-\tau R(d)}. \quad (9)$$

### III. EXPERIMENTAL METHODS

A single  $^{90}\text{Sr}$  source was used throughout, aligned to the centre of the Si detector window and placed behind a fixed plastic shield. The zero separation was defined as the minimum source-detector gap. Distances  $d$  (up to 90 cm) were then set by moving the stand along a ruler. At each  $d$ , the SimpleCount recorded counts  $n$  in a fixed sampling time  $\Delta t$ ; we calculated  $u(d)$  to test for deviations from a constant. Radioactive counting is Poisson with  $\sigma_n = \sqrt{n}$ , and the uncertainty in  $\Delta t$  is neglected. For each data point, we targeted the same percentage uncertainty in  $u(d)$ . In practice, we took one or two quick trial measurements to estimate  $R(d)$  and required  $\Delta t$  so that all points had comparable percentage statistical error.

After collecting count-rate data in 10 cm increments, the null model (constant  $u(d)$ ) was tested from the plotting of  $u(d)$  against  $d$ . We fitted a weighted constant to candidate distance windows and identified the region 16-60 cm, with  $\chi^2_{red} \approx 1.7$ , where the null model is most valid. We then took additional measurements near the end of this range with a finer spacing of 1 cm. Particularly in the near-field 0-10 cm, we collected data with a spacing of 1cm to better study deviations from the constant  $u(d)$ . The assumption that background count rate was negligible was tested by removing the source and recording counts for 2000 s. We obtained a background rate of  $R_{bg} = 0.12 \pm 0.01 \text{ s}^{-1}$ , which corresponds to  $\approx 0.7\%$  correction to the lowest count rate ( $\approx 14 \text{ s}^{-1}$ ). Therefore,  $R_{bg}$  was omitted from the corrections.

Dead time was obtained from five oscilloscope pulse traces by estimating the time duration in Python. Using a  $k\sigma$  threshold relative to the voltage baseline ( $\sigma$ : baseline noise;  $k$ : threshold multiplier). The estimated dead time is  $\tau = 3.12 \pm 0.06 \mu\text{s}$  with uncertainty from the scatter of the five measurements. The dead time correction is important only at small separations (e.g., a  $\sim 6.6 \pm 0.13\%$  correction at  $R(d) \sim 1.9 \times 10^4$ ); and negligible in the far field with a correction  $< 0.1\%$ .

### IV. RESULTS, ERRORS AND DISCUSSION

We first combined repeated measurements at each source-detector separation using weighted means in Fig. 1. The distance uncertainty was taken as a standard uncertainty derived from the ruler resolution,  $\sigma_d = 0.5/\sqrt{3} \text{ cm}$  (uniform rounding within  $\pm 0.5 \text{ cm}$ ). Using Eq. (9), we fixed the dead time and air density at  $\tau = 3.12 \pm 0.06 \mu\text{s}$  and  $\rho_{\text{air}} = 1.2 \times 10^{-3} \text{ g cm}^{-3}$ , and fitted the nuisance parameters  $d_0$ ,  $f_{\text{low}}$ , and  $K$ , obtaining ( $d_0 = 1.336 \pm 0.075 \text{ cm}$ ,  $f_{\text{low}} = 0.169 \pm 0.016$ , and  $K = 151300 \pm 2200 \text{ cps cm}^2$  for Model 2. The  $p$ -value for Model 2 is 0.87, indicating a good fit. The  $\Delta\chi^2_{red} \approx 33$  and  $\Delta\text{AIC}^6$  from Model 1  $\approx 178$ , showing a significant improvement. We can then obtain systematically corrected data. A weighted mean from Fig. 2 gives  $K_{est} = 161280 \pm 580 \text{ cps} \cdot \text{cm}^2$ , and the fitted WLS trend<sup>7</sup> is approximately flat, with a  $p$ -value 0.83, indicating that any remaining distance dependence is small compared with statistical uncertainties.

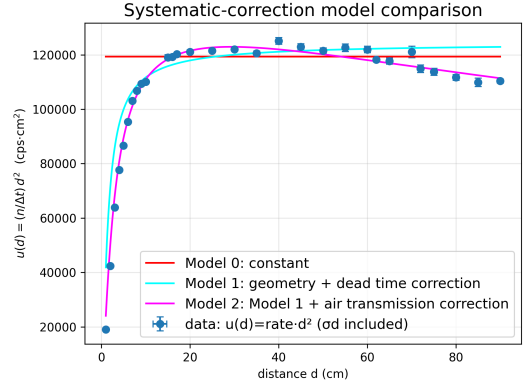


Fig. 1. Inverse-square-scaled rate  $u(d)$  versus distance with statistical and distance-propagated uncertainties. Model 1 reproduces the strong near-field deviation at small  $d$ , where the finite detector solid angle and dead-time losses dominate. Adding an air-transmission term (Model 2) further improves agreement in the far field by capturing the gradual reduction in rate with increasing air path length.

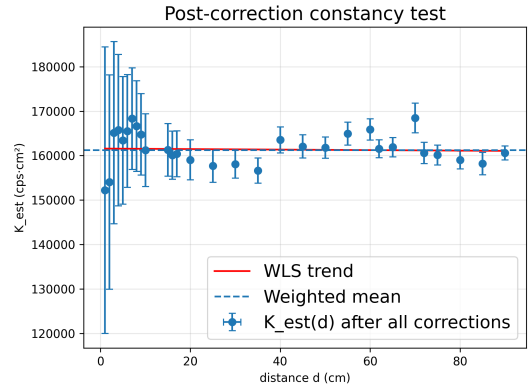


Fig. 2. After applying all corrections, the inferred parameter  $K_{est}(d)$  with  $\sigma_K$  from correction of systematic errors is consistent with a constant within uncertainties over most of the distance range.

Using Eq. 3 and  $K_{est}$ , we can obtain  $A\eta = (2.027 \pm 0.007) \times 10^6 \text{ Bq}$ . From the source label, the claimed activity of  $A$  is  $5 \times 10^6 \text{ Bq}$ . Therefore, the detection efficiency  $\eta$  is  $0.405 \pm 0.001$ .

### V. CONCLUSION

We measured the  $^{90}\text{Sr}$  count rate over  $d \simeq 0\text{-}90 \text{ cm}$  and analysed  $u(d)$ , which strongly deviated from a constant. The near-field deviation is mainly due to finite detector acceptance (rectangular solid angle) and a distance offset, with dead-time losses relevant only at high rates. The far-field deviation requires an air-transmission term, and the background count was negligible relative to the source signal. Using  $\tau = 3.12 \pm 0.06 \mu\text{s}$ ,  $\rho_{\text{air}} = 1.2 \times 10^{-3} \text{ g cm}^{-3}$  and fitting the full systematic model gave  $d_0 = 1.336 \pm 0.075 \text{ cm}$  and  $f_{\text{low}} = 0.169 \pm 0.016$ , with a good consistency ( $p \simeq 0.87$ ). After applying all systematic corrections,  $K_{est}(d)$  is consistent with a constant, producing  $K_{est} = 161280 \pm 580 \text{ cps} \cdot \text{cm}^2$  and an approximately flat WLS trend ( $p \simeq 0.83$ ). This implies  $A\eta = (2.027 \pm 0.007) \times 10^6 \text{ Bq}$  and  $\eta \simeq 0.405$  relative to the 5 MBq source label.

## APPENDIX

A. RECTANGULAR SOLID ANGLE<sup>3</sup>

For a rectangular detector with half-sides  $a, b$  on axis at distance  $d$ ,

$$\Omega_{rect}(d) = 4 \arctan \left( \frac{ab}{d\sqrt{d^2 + a^2 + b^2}} \right), \quad (10)$$

with  $a = b = 0.5\text{cm}$  for our  $1\text{cm} \times 1\text{cm}$  square detector

B.  $\beta$  SPECTRUM<sup>4</sup>

$$T(r) = \frac{\int_{E_{min}}^{E_{max}} S(E) dE}{\int_0^{E_{max}} S(E) dE}, \quad (11)$$

Where  $S(E)$  is the energy spectrum<sup>4</sup> of  $\beta$  for  $^{90}\text{Sr}$ .

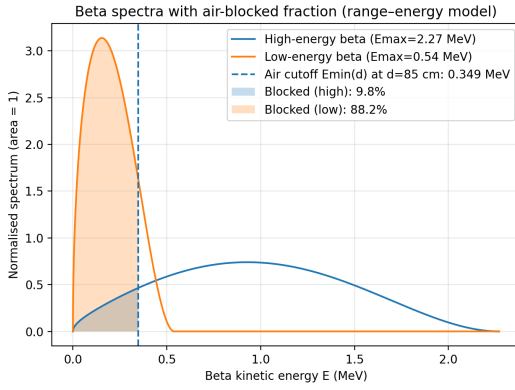


Fig. 3. Two different beta energy spectrums are blocked differently at distance 85cm

C. AIR AREAL DENSITY<sup>2</sup>

$$\sigma_{air}(r) = \rho_{air} r \quad (12)$$

## D. OBTAIN ROUGHLY THE SAME PERCENTAGE ERROR FOR DISTANCE

$$n_{req} \approx \frac{1}{\epsilon^2 - (2\sigma_d/d)^2}, \quad (13)$$

$$\Delta t(d) = \frac{n_{req}}{R_{trial}(d)}, \quad (14)$$

where  $\epsilon$  is the percentage error for  $u(d)$ .

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